BizBuddy

Case study 8: AI for inclusive MSME growth

A project by:

dawnwannahack

Team members:

Muhammad Adam Haikal bin Mohd Nor Azam (Leader)
Ajwad bin Mohd Tahrir
Nur Sakinah binti Muhammad Kamil Nasaruddin
Nurdeena Sofiya binti Zamzuri
Dawna Dowe Ajan

In the event of:

BorNEO HackWknd

dawnwannahack

BizBuddy

Introduction 03

Problem Statement and Objectives 04

Review of Existing Solution 05

Methodology 06

Results 07

Prototype Showcase 08

AI Acknowledgement 16

Conclusion 17

Introduction

Background of Case Study:

In Malaysia, micro-entrepreneurs like roadside food sellers and pasar tani farmers form the backbone of the local economy. However, they operate in a "shadow" economy—predominantly cash-based or using personal WhatsApp accounts for business. Because they lack formal financial records, they remain "invisible" to traditional banking systems, unable to secure the loans needed to scale.

Chosen SDGs:

We have chosen 3 SDGs that our project fulfills. NeuralCredit will help to improve our community, according to each goals.

SDG 1: No Poverty
By providing access to credit, we help families break the poverty cycle.

SDG 8: Decent Work & Economic Growth
Empowering micro-businesses to expand and hire.

SDG 9: Industry, Innovation, and Infrastructure
Bridging the digital divide through AI-driven financial infrastructure.

Problem Statement and Objectives

Problem Statement

While MSMEs are the backbone of the ASEAN economy, bank lending to this sector remains disproportionately small at only 16.9% of total lending (ADB, 2025). This is compounded by rising Non-Performing Loan (NPL) ratios, which reached 7.6% in certain regional segments in 2024.

In the current 2026 financial climate, banks are retreating from 'information-poor' borrowers to mitigate risk. This leaves millions of micro-entrepreneurs—who are hardworking but 'credit invisible'—without access to capital. BizBuddy solves this by using AI to transform informal WhatsApp data into high-fidelity financial records, lowering the NPL risk for banks and providing a bridge to modern credit guarantee schemes.

Objectives:

1. **Lower the Barrier:**
   Create a "zero-download" financial record system.

2. **Automate Visibility:**
   Use AI to turn informal WhatsApp conversations into formal ledger entries.

3. **Bridge the Gap:**
   Translate informal data into a "Bank-Readable" Credit Readiness Score.

Review of Existing Solutions

Current Solutions and Competitors

The current financial landscape for MSMEs is dominated by traditional banking institutions and established credit bureaus like CTOS and CCRIS. While these entities provide the foundation for the formal economy, they remain largely inaccessible to micro-entrepreneurs. Banks typically demand extensive physical documentation and tangible collateral—requirements that most informal businesses cannot meet. Meanwhile, credit bureaus focus strictly on tracking existing debt and repayment history; they offer no mechanism to build a profile for "credit invisible" individuals who have no prior formal borrowing record.

On the digital front, a variety of SME bookkeeping apps, such as BukuKas, have emerged to digitize financial records. However, these tools often struggle with sustained adoption among micro-businesses. They typically operate as standalone platforms that require users to navigate entirely new interfaces and commit to a disciplined routine of manual data entry. For a busy entrepreneur, these apps represent an additional administrative burden rather than a seamless integration into their existing daily workflow.

Gaps and Limitations

A primary barrier to digital adoption is "download fatigue" and technical constraints. Many micro-entrepreneurs operate on devices with limited storage and are reluctant to download and learn complex, data-heavy applications. This is compounded by the "manual tax"—the high friction of manual entry. When an entrepreneur is busy serving customers, logging every sale is often the first task to be neglected. Once a few transactions are missed, the records become incomplete, rendering the data useless for future credit assessments.

Furthermore, a significant language and accessibility barrier exists within the current fintech ecosystem. Most sophisticated financial tools are designed with an English-centric approach, which alienates the rural, BM-speaking demographic. By failing to offer localized, intuitive interfaces that speak the language of the user, these solutions remain disconnected from the very people who need them most. Ultimately, these gaps leave a massive segment of the MSME market financially stagnant and underserved.

Methodology

Tech Stack & Architecture

BizBuddy utilizes a high-performance, 4-layer architecture optimized for the 2026 ASEAN financial landscape. The Backend is powered by Python (Flask) on Railway, ensuring seamless auto-scaling for concurrent webhook requests. At its core, Google Gemini 2.5 Flash serves as the multimodal AI engine for instant receipt OCR and bilingual coaching. Data is managed via Firebase Firestore, enabling real-time synchronization between the WhatsApp bot and the Frontend PWA. We opted for a Vanilla JavaScript (ES6+) Frontend to eliminate framework overhead, resulting in a 200KB footprint that loads in <1s. This "stateless" and "lightweight" design ensures the system scales to 100k+ users while maintaining sub-3-second response times, even on the unstable 3G networks typical in rural MSME hubs.

Our methodology prioritized accessibility over complexity. By choosing a WhatsApp-first interface over a native mobile app, we eliminated download friction for 2.78 billion users, allowing record-keeping to happen within a familiar chat environment. To bridge the credit gap, we adopted a Hybrid Scoring Model: using deterministic, rule-based logic for financial accuracy and Gemini AI for natural language explanations. This dual approach ensures "Bank-Ready" transparency and regulatory compliance. The entire ecosystem is secured via a CI/CD pipeline through GitHub, Railway, and Netlify, providing an offline-first, resilient solution that turns informal business activity into a verifiable digital asset.

Implementation Process

The project was executed over a 10-day sprint using an agile, iterative methodology. Development was divided into four distinct phases :

1. Foundation: Establishing the WhatsApp webhook pipeline and a deterministic 6-criteria credit scoring algorithm.
2. AI Integration: Engineering prompts for Gemini to handle unstructured data parsing and bilingual business logic.
3. Frontend Excellence: Building a "zero-framework" Vanilla JS PWA to ensure maximum speed and offline-first functionality via Service Workers.
4. Multi-Country Validation: Implementing a COUNTRY_CONFIG architecture from Day 1, allowing the platform to support Malaysia, Indonesia, and the Philippines with zero code changes.

Results

Achievements of Objectives : Lower the Barrier

We successfully deployed a "zero-download" financial ecosystem by leveraging WhatsApp as the primary data entry point. As evidenced by the bot interface, users can record sales, generate credit scores, and receive AI business advice through a familiar chat environment. This eliminates the friction of traditional app onboarding, allowing micro-entrepreneurs to build a digital footprint without leaving their most-used communication tool. By meeting users where they are, we have ensured that even the most "tech-invisible" entrepreneurs can maintain consistent financial records.

Achievements of Objectives : Automate Visibility

Our second objective was met through the development of an AI-powered parsing engine that transforms unstructured WhatsApp conversations into formal, structured ledger entries. By automating the categorization of informal text (e.g., "sold 2 karipap for RM1") into standardized accounting headers, we have created a "living" financial record. This automation removes the burden of manual bookkeeping for the entrepreneur while providing real-time transparency. The outcome is a Progressive Web App (PWA) that reflects an accurate, up-to-the-minute financial health status, ensuring that business activities are no longer invisible to the formal economy.

Achievements of Objectives : Bridge the Gap

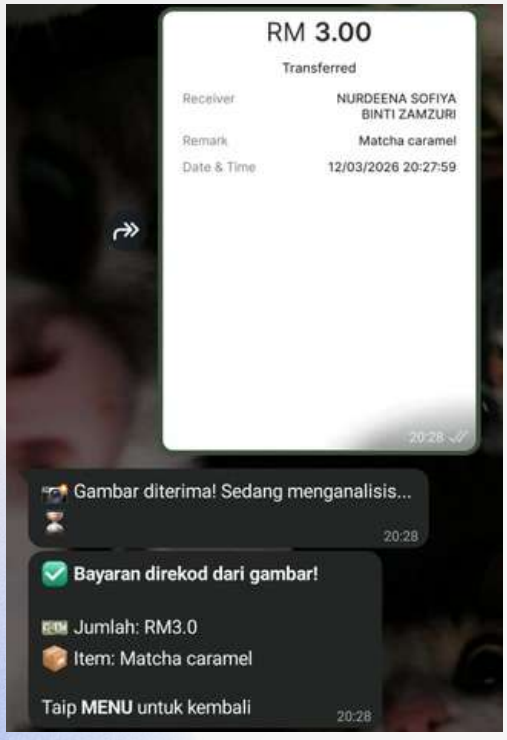
Finally, we bridged the gap between MSMEs and lenders by translating this informal data into a "Bank-Readable" Credit Readiness Score. By mapping consistent cash-flow patterns and behavioral data against regional risk metrics, BizBuddy provides banks with the granular visibility they need to lend confidently in a 2026 climate of 7.6% NPL ratios. We have successfully demonstrated that "credit-invisible" does not mean "high-risk." Our results confirm that by providing a standardized, data-rich bridge, we can turn hardworking micro-entrepreneurs into bankable assets, directly addressing the regional capital disparity.

Prototype Showcase

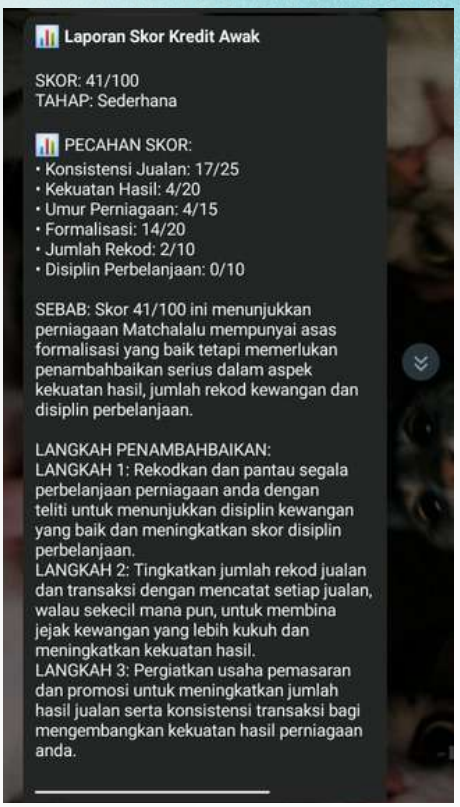
PART A : WhatsApp Bot

Upon completion of registration, type “menu” to view list of features

MENU 1 : Record sales by typing



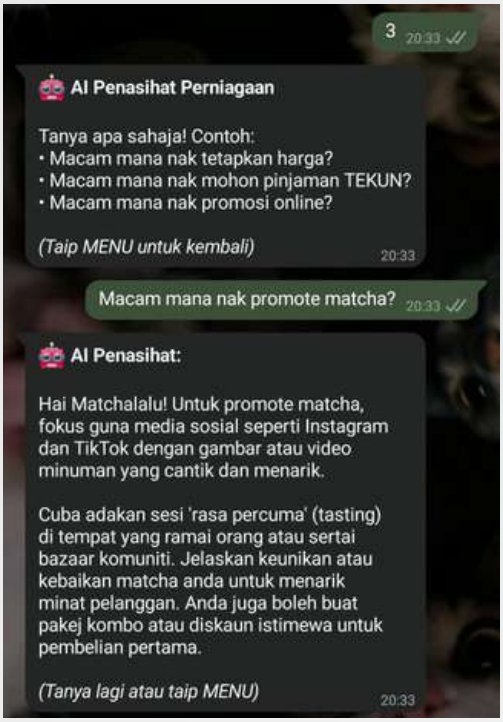
MENU 1 : Record sales by uploading picture



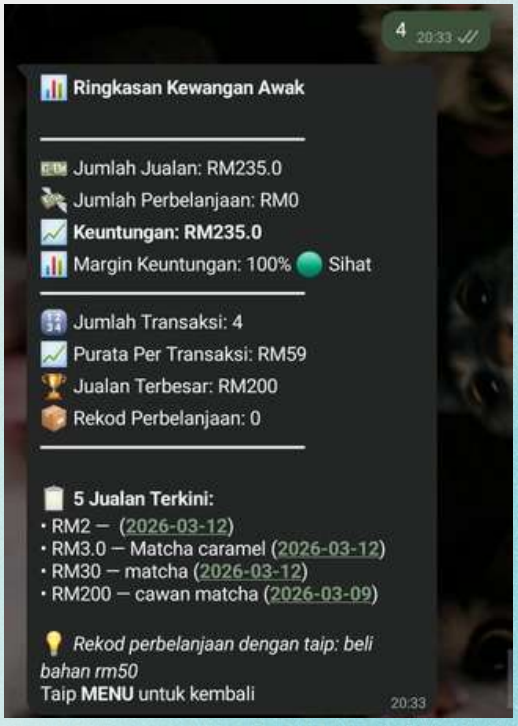
MENU 2 : Generate credit score report

Prototype Showcase

PART A : WhatsApp Bot

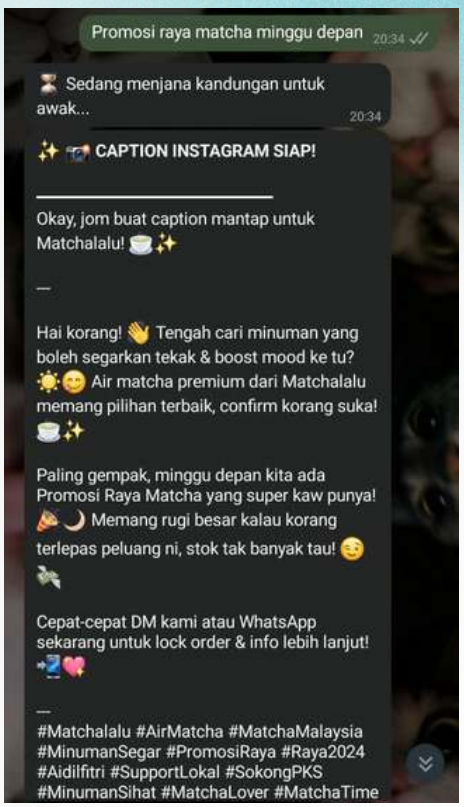


MENU 3 : Ask AI on any concern regarding your small business



MENU 4 : Financial summary

MENU 5 : Generate caption or content script for your social media



MENU 5 : Sample result for Instagram caption

Prototype Showcase

PART A : WhatsApp Bot

MENU 6 : AI sales forecast

MENU 7 : AI supply chain tips

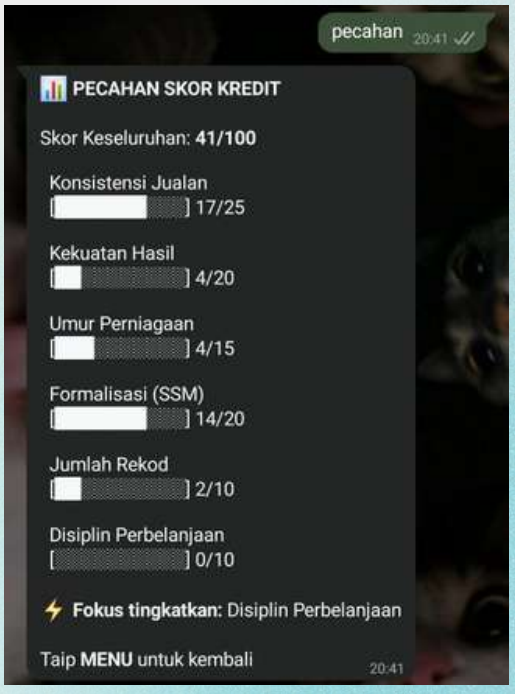
MENU 8 : ASEAN export guide

type “profile” to view saved business info

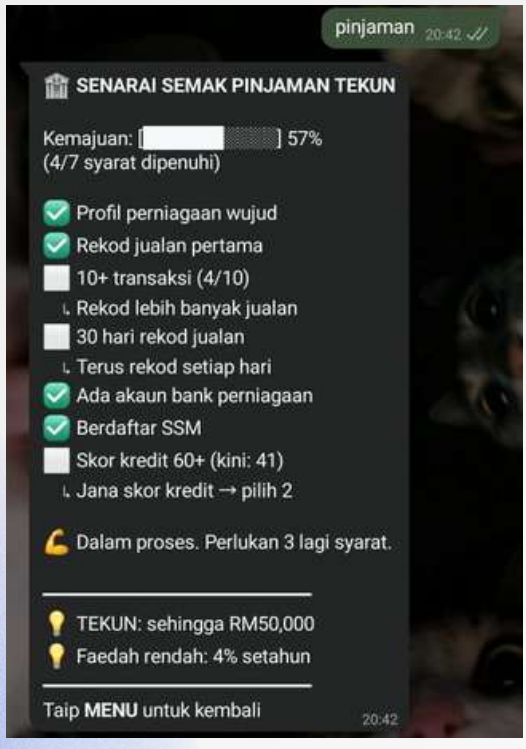
Prototype Showcase

PART A : WhatsApp Bot

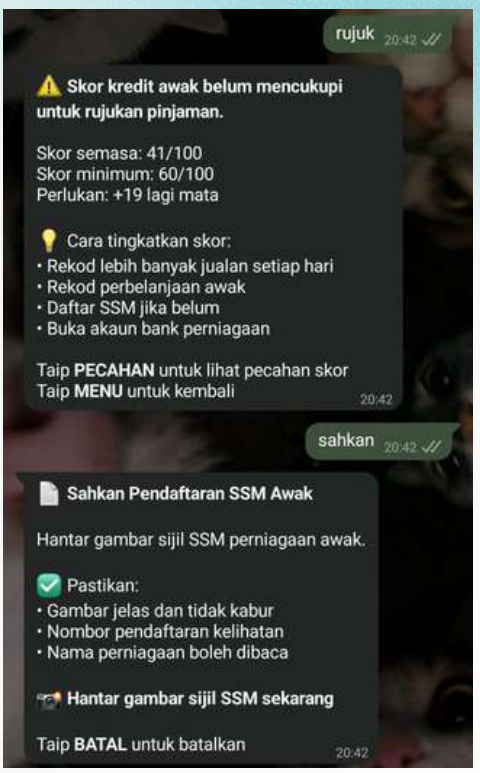
type “certificate” to generate credit readiness certificate



type “breakdown” to view credit score breakdown



type “loan” to view loan readiness checklist

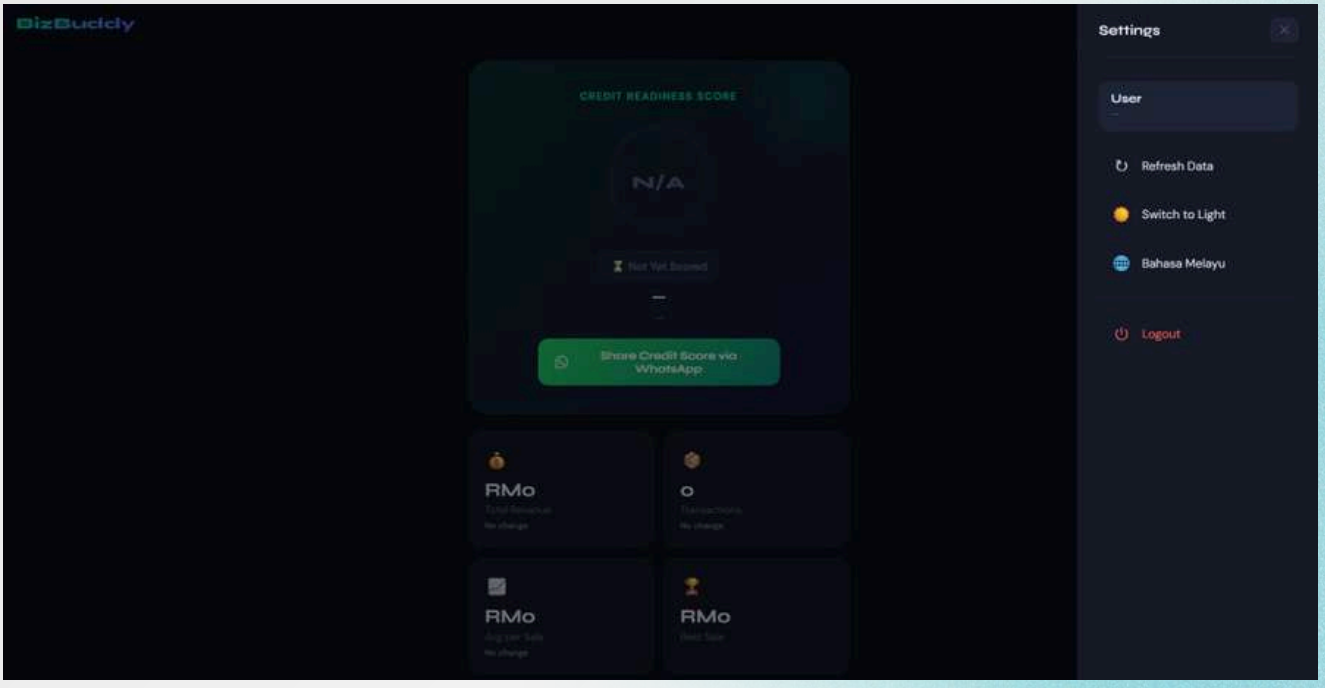


type “refer”to view loan referral eligibility and type “verify”to verify your business by uploading SSM certificate

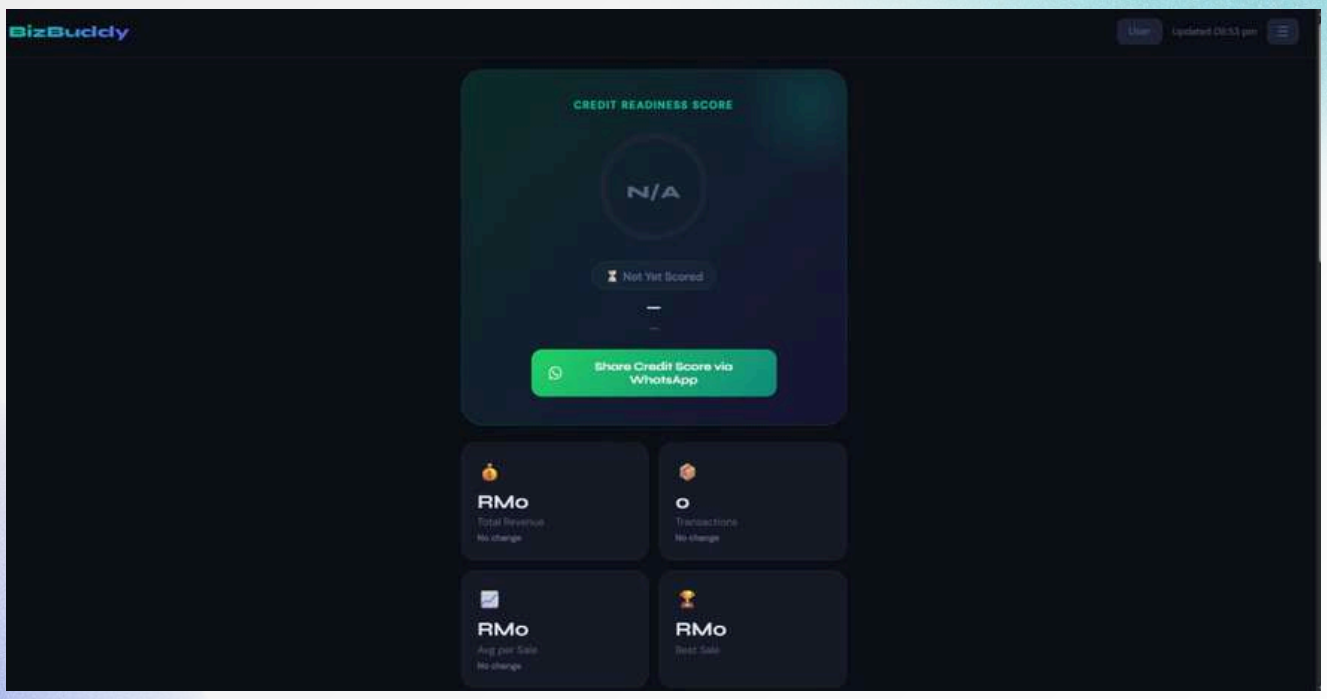
Prototype Showcase

PART B : BizBuddy Website

- **User Financial Profile**



Sidebar



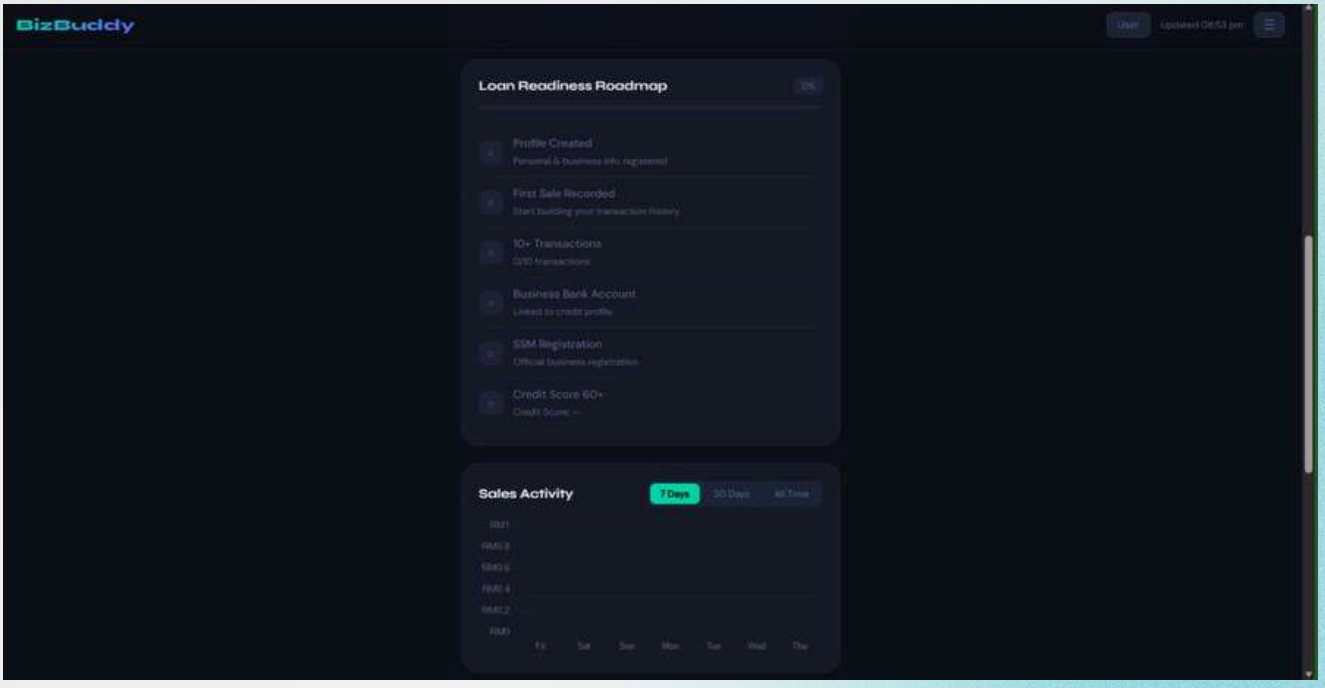
User dashboard

** All data in the website is synced with the data saved from the conversation with WhatsApp bot **

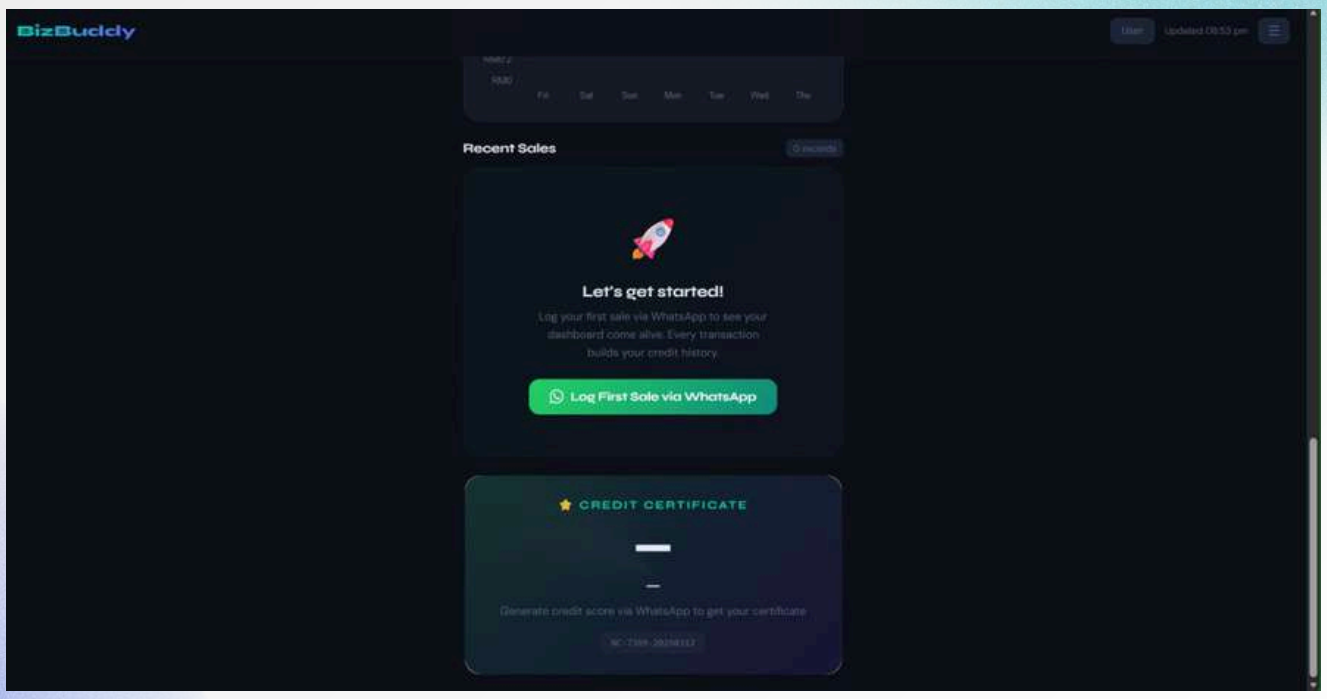
Prototype Showcase

PART B : BizBuddy Website

- **User Financial Profile**



Loan readiness roadmap and Sales activity



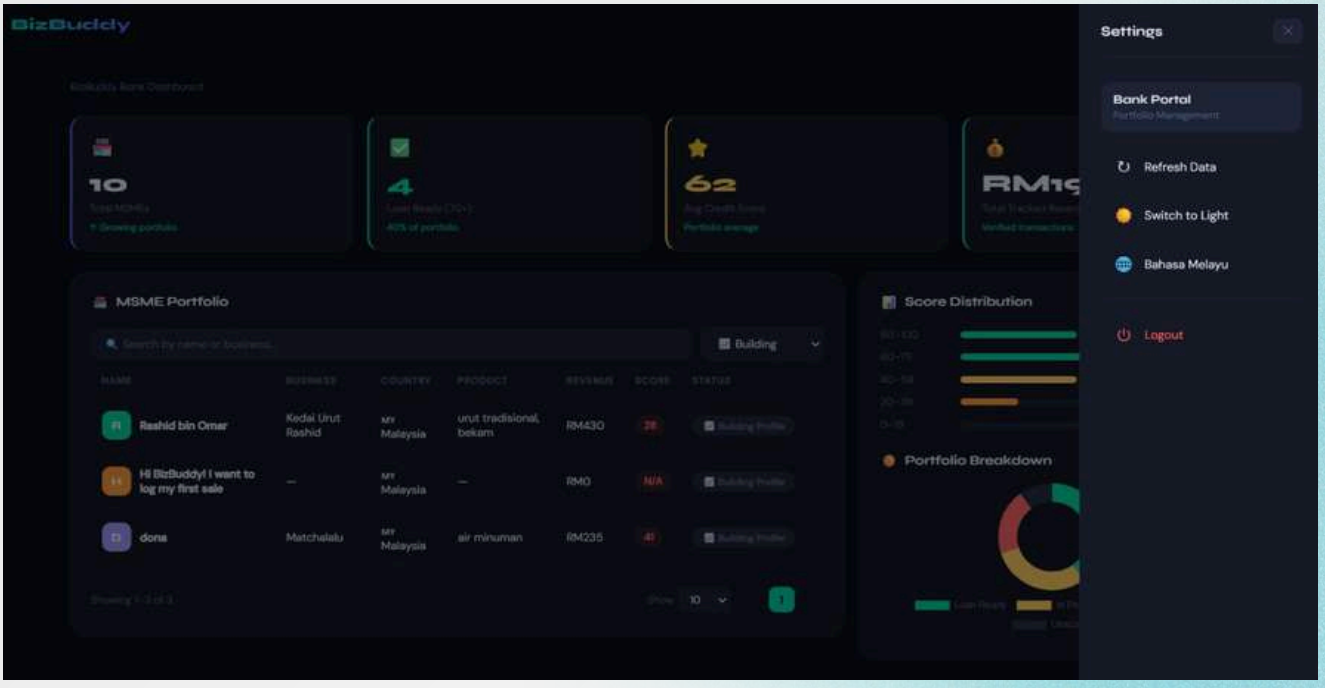
Recent sales and Credit score certificate

** All data in the website is synced with the data saved from the conversation with WhatsApp bot **

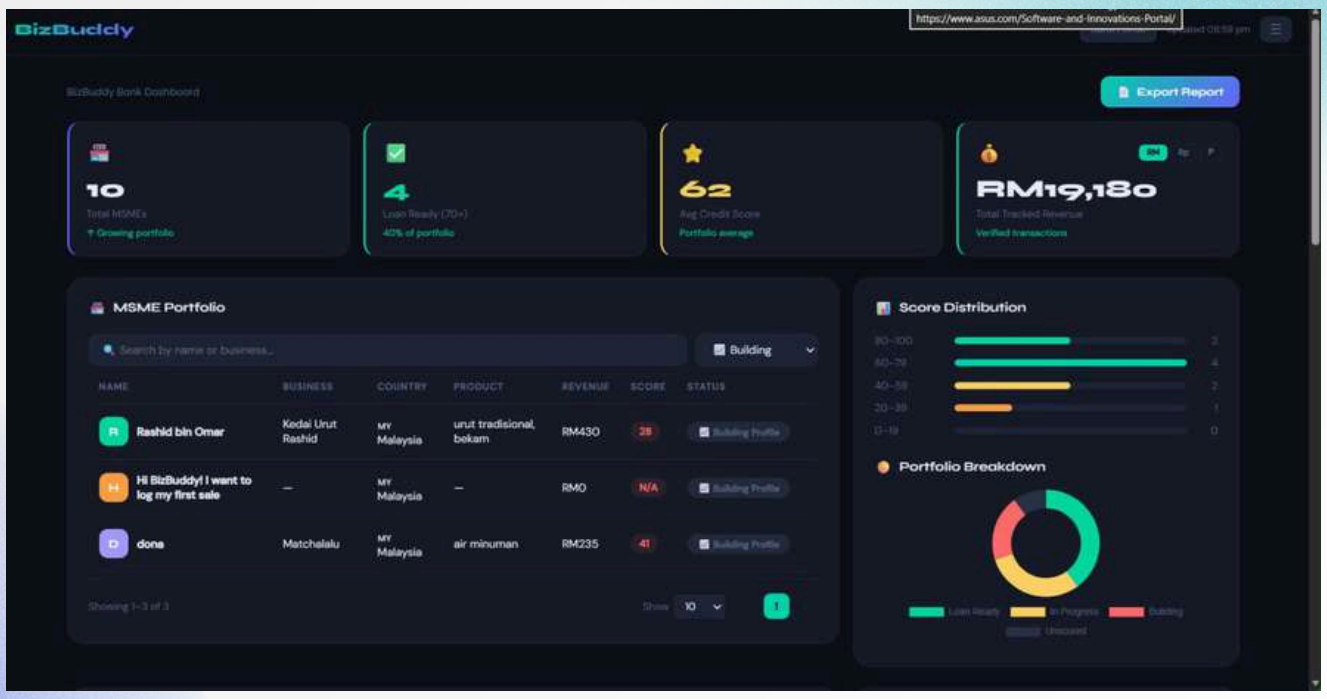
Prototype Showcase

PART B : BizBuddy Website

- **Bank View Portfolio Analytics**



Sidebar



Bank dashboard

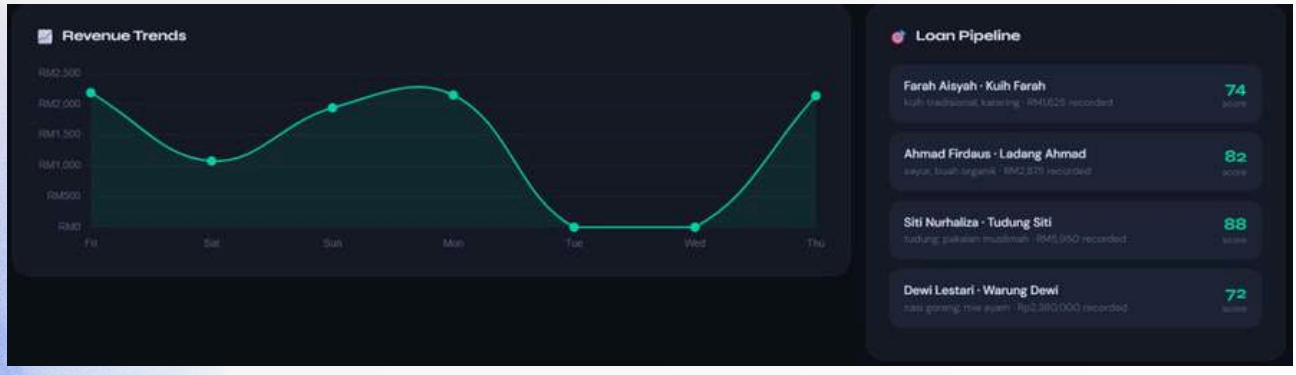
Prototype Showcase

PART B : BizBuddy Website

- **Bank View Portfolio Analytics**

Impact metrics

MSME distribution map in ASEAN (registered business)



Revenue trends and Loan pipeline

AI Acknowledgement

Development Process

In the development of BizBuddy, we utilized a multi-LLM approach to accelerate the build lifecycle and ensure technical rigor. **Claude** (Anthropic) served as our primary development partner for architectural design, codebase tuning, and UI/UX conceptualization. Following the initial build, **ChatGPT** (OpenAI) and **Gemini** (Google) were employed as secondary evaluators to stress-test our logic, rate feature ideation, and perform cross-platform peer reviews. This iterative feedback loop ensured a robust, thoroughly vetted prototype.

AI in solution

Beyond development, **Google Gemini 2.5 Flash** is integrated as a core component of the BizBuddy ecosystem. It functions as the primary "Intelligence Layer" within our WhatsApp bot, providing :

- Multimodal Vision : Powering our automated receipt OCR to extract financial data from user-uploaded images.
- Natural Language Processing : Converting unstructured chat messages into structured ledger entries and generating bilingual "Credit Readiness" explanations.
- Business Coaching : Providing real-time, context-aware business advice, revenue forecasting, and export guidance tailored to each MSME's unique cash-flow data.

Conclusion

Summary

BizBuddy represents a paradigm shift in micro-finance, moving beyond traditional banking hurdles to create an invisible layer of financial empowerment. By leveraging WhatsApp as the primary interface, the platform meets MSME owners exactly where they already operate, effectively removing the technical and psychological barriers that often keep small-scale entrepreneurs in the shadows of the formal economy. Our solution transforms the "everyday hustle" into a verifiable financial identity, proving that a mobile chat history can be just as powerful as a traditional credit score.

As we look toward the horizon, our goal is to further reduce friction and broaden our impact through strategic technological enhancements:

- **Multimodal Inputs:** We plan to implement Voice-to-Record functionality, allowing busy vendors to log transactions via simple voice notes. This ensures that even in the heat of a lunch rush, no sale goes unrecorded.
- **Regional Localization:** To scale our impact across Southeast Asia, we aim to localize our LLMs for Indonesian and Tagalog. This will provide tailored support for Warung owners and Sari-sari store operators, making the AI feel like a local partner rather than a foreign tool.
- **Seamless Financial Ecosystems:** By building Direct Bank Integrations, we will enable one-click loan applications within the WhatsApp thread. This bridge between informal data and formal capital will complete the loop of financial inclusion, turning daily records into immediate growth opportunities.

dawnwannahack

Thank You!

Our Project Information

GitHub Repo:
https://github.com/deena-org/dawnwannahack.git